

TEMPLATE P52 · ROOT CAUSE ANALYSIS AND CORRECTIVE ACTION PLAN

# Root cause analysis and corrective action plan

Worksheet for finding the systemic cause of a nonconformity or an AI incident, analysing it without seeking to apportion blame and turning the conclusion into actions with an owner, a time limit and an effectiveness criterion.

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**Why it matters.** Correcting the event without eliminating its cause guarantees that it will recur, in the same system or in another that shares a supplier, model or process. An analysis that ends in "human error" protects the management system and exposes people, and means that next time nobody reports in time. This worksheet sets the method, requires the root cause to meet four validity conditions and forces a decision on how the effectiveness of each action will be measured before it is executed; it is the evidence the AI Auditor needs to close a nonconformity (P50) and the post-incident analysis that closes an incident (P27).

# Instructions for use

| Aspect                         | Guidance                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>When it is used</b>         | Mandatory for major and critical nonconformities and for the post-incident analysis of S1, S2 and S3 incidents (simplified for S3); recommended for recurring minor nonconformities. Time limits: 37 §3.10 (nonconformities) and 37 §4.3 (post-incident analysis: 10 working days after technical closure for S1, 20 for S2, 30 days for S3). |
| <b>Who completes it</b>        | The action owner (nonconformities) or the incident coordinator (incidents), with the technical team and the people who were involved. A facilitator from outside the team is advisable for S1 and critical nonconformities.                                                                                                                   |
| <b>Who verifies it</b>         | The AI Risk Owner reviews the post-incident analysis of incidents (37 §4.5). The AI Auditor checks the validity of the root cause and the effectiveness before closing a nonconformity.                                                                                                                                                       |
| <b>Who decides or approves</b> | The plan is approved by the AI Committee (critical and S1), the AI Risk Owner (major and S2) or the AI Office (minor and S3), in accordance with 37 §3.6.                                                                                                                                                                                     |
| <b>Stage, gate or moment</b>   | Phase 6 and any time a nonconformity is opened. It is reviewed at R6 (R6.06, R6.15), at G7 (G7.04) and in the C5 annual review (recurrences).                                                                                                                                                                                                 |
| <b>Tool</b>                    | T08 · Nonconformity and incident register (actions and status). The worksheet is kept as linked evidence.                                                                                                                                                                                                                                     |
| <b>Reference document</b>      | Document 37 §2 (principles 4 and 6), §3.5–3.7 and §4.5 (phases 7 and 8); document 33 (materialised risks); document 35 (controls).                                                                                                                                                                                                            |
| <b>Lite versus Enterprise</b>  | Lite may use only the five whys and omit the blocks marked <b>(Enterprise)</b> . The validation of the root cause and the effectiveness criterion are mandatory in both.                                                                                                                                                                      |

## Completion rules:

- **No blame, but clear accountability.** Causes are sought in the management system, not people to whom the failure can be attributed. "Human error" is not a final root cause: the question is why the system allowed that error or failed to detect it.
- Before the analysis starts, it is checked that the evidence from the time of the incident has been preserved (37 §2, principle 7).
- Several nonconformities with the same cause are analysed in a single worksheet, but each keeps its own code (37 §3.3).
- Every action considers whether the same failure exists in other systems or initiatives (the "correct here and everywhere" principle).

## Identification

| Field                   | Content                                                                   | Guidance |
|-------------------------|---------------------------------------------------------------------------|----------|
| Subject of the analysis | NC-AAAA-NNN · INC-AAAA-NNN (one or several).                              |          |
| Type or severity        | Minor · Major · Critical / S1 · S2 · S3 · S4.                             |          |
| System and initiative   | SIA-AAAA-NNN · IA-AAAA-NNN.                                               |          |
| Facilitator             | Role and name; from outside the team where advisable.                     |          |
| Participants            | Roles, not just names.                                                    |          |
| Start date and deadline | In accordance with 37 §3.10 or §4.3.                                      |          |
| Evidence preserved      | Logs, model and instruction versions, configurations, inputs and outputs. |          |
| Worksheet version       |                                                                           |          |

## Facts and timeline

Verifiable facts are described, in order, without interpretation.

| Date and time                           | Fact                                                  | Source or evidence  | Who or what acted                       |
|-----------------------------------------|-------------------------------------------------------|---------------------|-----------------------------------------|
| (illustrative example) 03-11-2026 09:12 | The supplier's model version is replaced by a new one | Supplier change log | Supplier, without notifying the company |
| (illustrative example) 03-11-2026 15:40 | The rejection rate alert exceeds its threshold        | Alert AL-04 in P25  | Monitoring                              |

| Field           | Content                           | Guidance |
|-----------------|-----------------------------------|----------|
| Immediate cause | What directly produced the event. |          |

| Field             | Content | Guidance                                                   |
|-------------------|---------|------------------------------------------------------------|
| What worked       |         | Controls that detected or limited the harm.                |
| What did not work |         | Controls that failed, did not exist or were not triggered. |

## 04

# Five whys

The question of why each answer occurred is asked until a cause is reached that can be acted upon and that explains the failure of the management system, not just the event. More or fewer than five steps may be needed.

| Step | Question                 | Answer | Evidence |
|------|--------------------------|--------|----------|
| 1    | Why did the event occur? |        |          |
| 2    | Why did the above occur? |        |          |
| 3    | Why?                     |        |          |
| 4    | Why?                     |        |          |
| 5    | Why?                     |        |          |

*(illustrative example)* The assistant gave incorrect return periods → because the supplier changed the model version → because the contract did not require advance notice of version changes → because the supplier assessment did not review that clause → because P14 did not include it for N2 suppliers. Root cause: the list of clauses applied to N2 suppliers did not require advance notice of model changes.

## 05

# Cause-and-effect diagram (Ishikawa) (Enterprise)

Possible causes are explored by category using the guiding questions in 37 §3.5. Only causes for which there are indications are recorded; those ruled out are marked with the reason.

| Category                       | Guiding questions                                                                                                | Possible causes | Evidence | Confirmed? (Yes · No · Under analysis) |
|--------------------------------|------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------------------------------------|
| <b>People and capabilities</b> | Were they aware of the requirement? Did they have the training and the time? Was there pressure to move forward? |                 |          |                                        |
| <b>Process and method</b>      | Was the requirement clear? Did the template or checklist include it? Did the workflow allow it to be skipped?    |                 |          |                                        |
| <b>Data</b>                    | Did the data, its quality or its origin change? Was there lineage?                                               |                 |          |                                        |
| <b>Model and technology</b>    | Did the model, the instructions or the configuration change? Did monitoring fail?                                |                 |          |                                        |
| <b>Suppliers</b>               | Were there unreported changes by the supplier? Did the contract cover this?                                      |                 |          |                                        |
| <b>Governance and controls</b> | Was there segregation of duties? Was the control designed and tested? Did the register raise an alert?           |                 |          |                                        |
| <b>Environment</b>             | Did the regulation, the use or the volume change, or did an attacker appear?                                     |                 |          |                                        |

## 06

# Validation of the root cause

| Condition (37 §3.5)                                                                      | Met? (Yes · No) | Justification |
|------------------------------------------------------------------------------------------|-----------------|---------------|
| It explains all the facts observed.                                                      |                 |               |
| Had it been eliminated, the failure would not have occurred or would have been detected. |                 |               |
| It is actionable.                                                                        |                 |               |
| It does not merely attribute responsibility to a person.                                 |                 |               |

| Field                                                  | Content | Guidance                                                                                                            |
|--------------------------------------------------------|---------|---------------------------------------------------------------------------------------------------------------------|
| Confirmed root cause                                   |         | One sentence. If there are several, they are numbered.                                                              |
| Was the risk in the register?                          |         | Yes (IA-AAAA-NNN · <i>Rnn</i> , moves to <i>Materialised</i> and is reassessed, 33 §7.2) · No (it is added to P12). |
| Other systems or initiatives exposed to the same cause |         | Same supplier, model, data, template or process.                                                                    |

## 07

# Blameless post-incident analysis

For incidents only (37 §4.5, phase 7). It is completed in addition to sections 3 to 6.

| Aspect                                              | Content | Guidance                                                                                                                 |
|-----------------------------------------------------|---------|--------------------------------------------------------------------------------------------------------------------------|
| Effectiveness of detection                          |         | Hours between estimated start and detection; what would have brought it forward.                                         |
| Effectiveness of containment                        |         | Hours between detection and containment; was the kill switch used and how long did it take to take effect?               |
| Compliance with internal and regulatory time limits |         | In accordance with P51 §10.                                                                                              |
| Harm and cost                                       |         | Amount and status (validated · declared · estimated), or <i>No data</i> .                                                |
| Effectiveness of controls                           |         | SEG and AG controls that worked or failed.                                                                               |
| Decisions taken under pressure                      |         | What information was missing and how the decision was made; without judging the people involved.                         |
| Lessons for other initiatives (Enterprise)          |         | Initiatives with shared technology or suppliers to which they are communicated.                                          |
| Should nonconformities be opened?                   |         | If the incident reveals a requirement not met or if the management breached the process (37 §6); NC codes opened in P50. |

# Corrective and preventive action plan

**Type:** Correction (remedy the event) · Corrective action (eliminate the root cause) · Preventive action (prevent the same cause from producing the failure in other systems). **Status:** Pending · In progress · Executed · Verified effective · Not effective.

| No.                        | Cause addressed                    | Type              | Action                                                                                   | Owner                                | Area with which it is agreed | Time limit | Effectiveness criterion                                                  | Expected evidence               |
|----------------------------|------------------------------------|-------------------|------------------------------------------------------------------------------------------|--------------------------------------|------------------------------|------------|--------------------------------------------------------------------------|---------------------------------|
| A01                        |                                    |                   |                                                                                          |                                      |                              |            |                                                                          |                                 |
| A02                        |                                    |                   |                                                                                          |                                      |                              |            |                                                                          |                                 |
| A03                        |                                    |                   |                                                                                          |                                      |                              |            |                                                                          |                                 |
| (illustrative example) A01 | No advance notice of model changes | Corrective action | Add the advance notice clause to the N2 list in P14 and renegotiate the current contract | AI Technical Owner, with procurement | Legal counsel                | 31-01-2027 | Contract signed with the clause; next version change notified in advance | Signed addendum supplier notice |
| (illustrative example) A02 | Same cause at other suppliers      | Preventive action | Review the contracts of all N2 and N3 model suppliers                                    | AI Office                            | Procurement                  | 28-02-2027 | 100 % of contracts reviewed; those without the clause, with a plan       | Review li                       |

| Field             | Content | Guidance                                                                               |
|-------------------|---------|----------------------------------------------------------------------------------------|
| Plan approved by  |         | In accordance with 37 §3.6.                                                            |
| Approval date     |         | The effectiveness criterion is fixed on this date.                                     |
| Resulting updates |         | Risk register (P12), controls (P18), suppliers (P14), templates, test scenarios (P53). |

## Effectiveness verification

| Action no. | Verification date | Period or sample observed | Result (Effective · Not effective) | Evidence | Verified by |
|------------|-------------------|---------------------------|------------------------------------|----------|-------------|
|            |                   |                           |                                    |          |             |
|            |                   |                           |                                    |          |             |

If an action is not effective, the nonconformity returns to **Under analysis** (37 §3.7) and this worksheet is revised from section 6.

## Quality criteria

Formal criteria in document 21 (R6.06, R6.15 and G7.04) and method in 37 §3.5–3.7.

| # | Check                                                                                             | Status                                   |
|---|---------------------------------------------------------------------------------------------------|------------------------------------------|
| 1 | The timeline is based on evidence preserved from the time of the event.                           | Met · Not met · Not applicable · Pending |
| 2 | The immediate cause is distinguished from the root cause.                                         |                                          |
| 3 | The root cause meets the four validity conditions and does not end in "human error".              |                                          |
| 4 | It has been checked whether other systems or initiatives are exposed to the same cause.           |                                          |
| 5 | Each action has a type, an owner, a time limit, an effectiveness criterion and expected evidence. |                                          |
| 6 | There is at least one corrective action for each root cause; correction alone is not enough.      |                                          |
| 7 | The risk register and the affected controls have been updated.                                    |                                          |
| 8 | Effectiveness has been verified with the criterion set when the plan was approved.                |                                          |

# Approval and verification

| Function                           | Role in SEVEN-G                                                                         | Name and position | Date | Signature or reference |
|------------------------------------|-----------------------------------------------------------------------------------------|-------------------|------|------------------------|
| Prepares                           | Action owner or incident coordinator                                                    |                   |      |                        |
| Facilitates (Enterprise)           | Person from outside the team                                                            |                   |      |                        |
| Reviews the post-incident analysis | AI Risk Owner                                                                           |                   |      |                        |
| Approves the plan                  | AI Committee · AI Risk Owner · AI Office, according to type or severity                 |                   |      |                        |
| Verifies effectiveness             | AI Auditor (nonconformities) · AI Risk Owner (incident actions without a nonconformity) |                   |      |                        |

Segregation of duties: whoever gave rise to the event takes part by providing information, but does not validate the root cause or verify the effectiveness of the actions.

# Version control

## Completed document

| Version | Date | Author | Changes |
|---------|------|--------|---------|
|---------|------|--------|---------|

## Template

| Version | Date       | Changes                                                                                                                                                                                                     |
|---------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.1     | 19-09-2026 | First version. Five whys, Ishikawa with the AI categories of 37 §3.5, validation of the root cause, blameless post-incident analysis, corrective and preventive action plan and effectiveness verification. |